Лабораторная работа №1

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Цель работы

Проанализировать последовательность загрузки системы с помощью команды `dmesg` и `journalctl`

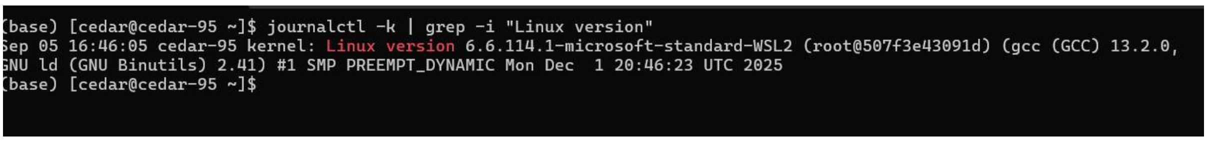
Команды для анализа

- `dmesg` — вывод сообщений ядра
- `journalctl -k` — просмотр журнала ядра
- `grep` — фильтрация вывода

Полученная информация

Параметр	Результат
Версия ядра	Linux version ...
Частота процессора	... MHz
Модель процессора	...
Оперативная память	... MB/GB
Тип гипервизора	...
Файловая система	...
Монтирование	...

Версия ядра Linux



```
(base) [cedar@cedar-95 ~]$ journalctl -k | grep -i "Linux version"
Sep 05 16:46:05 cedar-95 kernel: Linux version 6.6.114.1-microsoft-standard-WSL2 (root@507f3e43091d) (gcc (GCC) 13.2.0,
GNU ld (GNU Binutils) 2.41) #1 SMP PREEMPT_DYNAMIC Mon Dec  1 20:46:23 UTC 2025
(base) [cedar@cedar-95 ~]$
```

Процессор

```
(base) [cedar@cedar-95 ~]$ journalctl -k | grep -i "Mhz"
Sep 05 16:46:05 cedar-95 kernel: tsc: Detected 2611.213 MHz processor
```

```
(base) [cedar@cedar-95 ~]$ journalctl -k | grep -i "CPU"
Sep 05 16:46:05 cedar-95 kernel: Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=12 hv_utils.timesync_
implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
Sep 05 16:46:05 cedar-95 kernel: KERNEL supported cpus:
Sep 05 16:46:05 cedar-95 kernel: smpboot: Allowing 12 CPUs, 0 hotplug CPUs
Sep 05 16:46:05 cedar-95 kernel: setup_percpu: NR_CPUS:8192 nr_cpumask_bits:12 nr_cpu_ids:12 nr_node_ids:1
Sep 05 16:46:05 cedar-95 kernel: percpu: Embedded 63 pages/cpu s221184 r8192 d28672 u262144
Sep 05 16:46:05 cedar-95 kernel: pcpu-alloc: s221184 r8192 d28672 u262144 alloc=1*2097152
Sep 05 16:46:05 cedar-95 kernel: pcpu-alloc: [0] 00 01 02 03 04 05 06 07 [0] 08 09 10 11 -- -- -- --
Sep 05 16:46:05 cedar-95 kernel: Kernel command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=12 hv_utils.ti
mesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
Sep 05 16:46:05 cedar-95 kernel: SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=12, Nodes=1
Sep 05 16:46:05 cedar-95 kernel: rcu:         RCU restricting CPUs from NR_CPUS=8192 to nr_cpu_ids=12.
Sep 05 16:46:05 cedar-95 kernel: rcu: Adjusting geometry for rcu_fanout_leaf=16, nr_cpu_ids=12
Sep 05 16:46:05 cedar-95 kernel: x86/cpu: User Mode Instruction Prevention (UMIP) activated
Sep 05 16:46:05 cedar-95 kernel: smpboot: CPU0: 13th Gen Intel(R) Core(TM) i5-13420H (family: 0x6, model: 0xba, stepping
: 0x2)
Sep 05 16:46:05 cedar-95 kernel: RCU Tasks: Setting shift to 4 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=12.
Sep 05 16:46:05 cedar-95 kernel: RCU Tasks Rude: Setting shift to 4 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=1
2.
Sep 05 16:46:05 cedar-95 kernel: RCU Tasks Trace: Setting shift to 4 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=
12.
Sep 05 16:46:05 cedar-95 kernel: Performance Events: unsupported p6 CPU model 186 no PMU driver, software events only.
Sep 05 16:46:05 cedar-95 kernel: smp: Bringing up secondary CPUs ...
Sep 05 16:46:05 cedar-95 kernel: .... node  #0, CPUs:        #2  #4  #6  #8 #10  #1  #3  #5  #7  #9 #11
Sep 05 16:46:05 cedar-95 kernel: smp: Brought up 1 node, 12 CPUs
Sep 05 16:46:05 cedar-95 kernel: cpuidle: using governor ladder
Sep 05 16:46:05 cedar-95 kernel: cpuidle: using governor menu
Sep 05 16:46:05 cedar-95 kernel: cryptd: max_cpu_qlen set to 1000
Sep 05 16:46:05 cedar-95 kernel: ACPI: _OSC evaluation for CPUs failed, trying _PDC
Sep 05 16:46:05 cedar-95 kernel: intel_pstate: CPU model not supported
```

Оперативная память

```
(base) [cedar@cedar-95 ~]$ journalctl -k | grep -i "Memory"
Sep 05 16:46:05 cedar-95 kernel: ACPI: Reserving FACP table memory at [mem 0x101000-0x101113]
Sep 05 16:46:05 cedar-95 kernel: ACPI: Reserving DSDT table memory at [mem 0x1011b8-0x11f348]
Sep 05 16:46:05 cedar-95 kernel: ACPI: Reserving FACS table memory at [mem 0x101114-0x101153]
Sep 05 16:46:05 cedar-95 kernel: ACPI: Reserving OEM0 table memory at [mem 0x101154-0x1011b7]
Sep 05 16:46:05 cedar-95 kernel: ACPI: Reserving SRAT table memory at [mem 0x11f349-0x11f6b8]
Sep 05 16:46:05 cedar-95 kernel: ACPI: Reserving APIC table memory at [mem 0x11f6b9-0x11f760]
Sep 05 16:46:05 cedar-95 kernel: Early memory node ranges
Sep 05 16:46:05 cedar-95 kernel: PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
Sep 05 16:46:05 cedar-95 kernel: PM: hibernation: Registered nosave memory: [mem 0x000a0000-0x001fffff]
Sep 05 16:46:05 cedar-95 kernel: PM: hibernation: Registered nosave memory: [mem 0xf8000000-0xffffffff]
Sep 05 16:46:05 cedar-95 kernel: Memory: 4074488K/8245884K available (20480K kernel code, 3451K rwdata, 13596K rodata, 4
480K init, 6224K bss, 263532K reserved, 0K cma-reserved)
Sep 05 16:46:05 cedar-95 kernel: Freeing SMP alternatives memory: 44K
Sep 05 16:46:05 cedar-95 kernel: x86/mm: Memory block size: 128MB
Sep 05 16:46:05 cedar-95 kernel: Freeing initrd memory: 2772K
Sep 05 16:46:05 cedar-95 kernel: hv_balloon: Using Dynamic Memory protocol version 2.0
Sep 05 16:46:05 cedar-95 kernel: hv_balloon: Cold memory discard hint enabled with order 9
Sep 05 16:46:05 cedar-95 kernel: Freeing unused decrypted memory: 2028K
Sep 05 16:46:05 cedar-95 kernel: Freeing unused kernel image (initmem) memory: 4480K
Sep 05 16:46:05 cedar-95 kernel: Freeing unused kernel image (rodata/data gap) memory: 740K
Sep 05 16:46:50 cedar-95 kernel: hv_balloon: Max. dynamic memory size: 8054 MB
```

Файловая система и монтирование

```
(base) [cedar@cedar-95 ~]$ journalctl -k | grep -i "filesystem"
Sep 05 16:46:05 cedar-95 kernel: EXT4-fs (sda): mounted filesystem 00000000-0000-0000-0000-000000000000 ro without journ
al. Quota mode: none.
Sep 05 16:46:05 cedar-95 kernel: EXT4-fs (sdb): mounted filesystem 00000000-0000-0000-0000-000000000000 ro without journ
al. Quota mode: none.
Sep 05 16:46:05 cedar-95 kernel: EXT4-fs (sdd): mounted filesystem 4febe0b0-8d81-47b6-b832-23dd124cb4e0 r/w with ordered
 data mode. Quota mode: none.
```

mount

Выводы

1. Получена информация о загрузке системы
2. `dmesg` и `journalctl` — полезные инструменты
3. Навыки диагностики системы закреплены

Спасибо за внимание!